

COMPSCI 689

Lecture 7: Support Vector Machines

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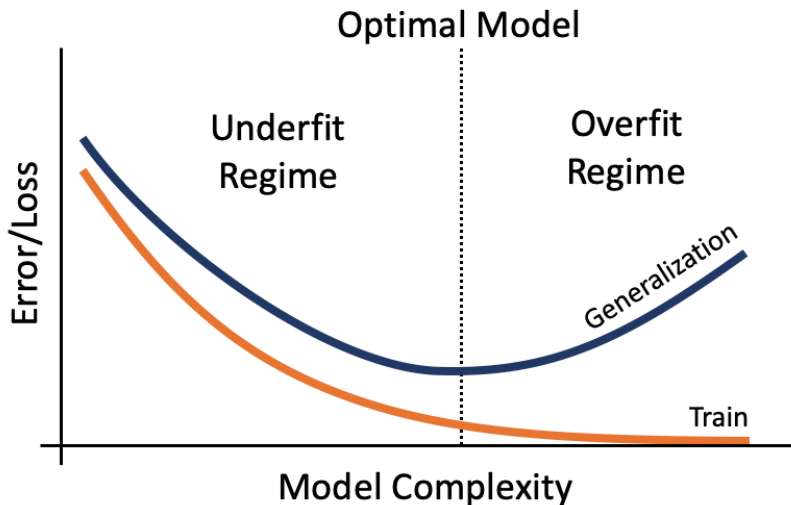
Basis Function Expansion

- A simple solution to the linearity problem is to apply a set of non-linear functions $\phi_1, \dots, \phi_K$ to the raw feature vector $\mathbf{x}$ to map it in to a new feature space:

$$\phi(\mathbf{x}) = [\phi_1(\mathbf{x}), \dots, \phi_K(\mathbf{x})]$$

- This is called a *basis function expansion* since $K > D$ in general. This requires that we know the functions $\phi_1, \dots, \phi_K$ that we want to apply in advance.
- We then define a linear model in this new feature space using a weight vector $\theta \in \mathbb{R}^K$.
- In the regression setting, we obtain the model $\hat{y} = \phi(\mathbf{x})\theta$.
- In the classification setting, we obtain $\hat{y} = \text{sign}(\phi(\mathbf{x})\theta)$.

Overfitting and Underfitting



Regularized Risk Minimization

- When regularization is applied in the expected risk minimization framework, the resulting framework is referred to as *regularized risk minimization* (RRM).
- Below, we show the RRM objective function for the case where the regularization function is based on the squared two norm:

$$\hat{\theta} = \arg \min_{\theta \in \Theta} \left(\left(\frac{1}{N} \sum_{n=1}^N L(y_n, f_{\theta}(\mathbf{x}_n)) \right) + \lambda \|\mathbf{w}\|_2^2 \right)$$

- $\lambda \geq 0$ is the regularization hyper-parameter.

Support Vector Machines

- Support vector machines are a class of supervised learning approaches based on linear models combined with convex, non-differentiable loss functions.
- There are specific approaches for both regression (SVR) and classification (SVC).
- The models are learned using regularized risk minimization and can be used with basis expansions.
- The models learned using these loss functions have some interesting properties relative to the squared loss and logistic loss.

Convex Functions and Sets

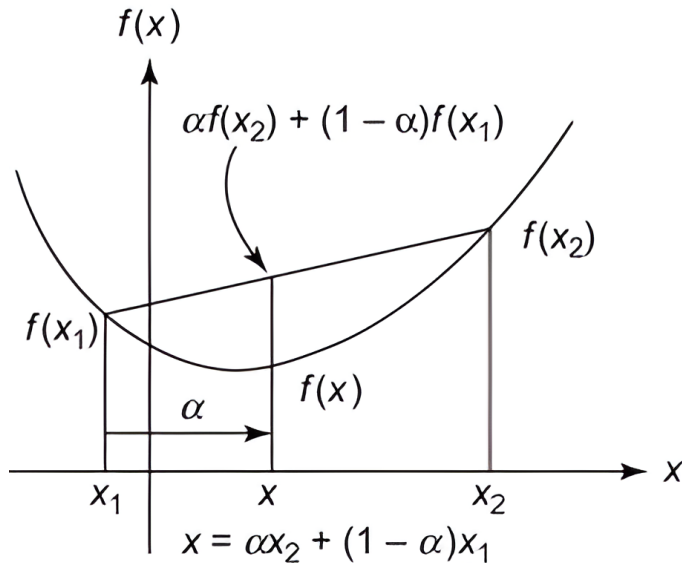
A function $f : \mathcal{X} \rightarrow \mathcal{R}$ is said to be convex if $\mathcal{X}$ is a convex set and for any two points $\mathbf{x}_1, \mathbf{x}_2 \in \mathcal{X}$ and any scalar $\alpha \in [0, 1]$:

$$f(\alpha \mathbf{x}_2 + (1 - \alpha) \mathbf{x}_1) \leq \alpha f(\mathbf{x}_2) + (1 - \alpha) f(\mathbf{x}_1)$$

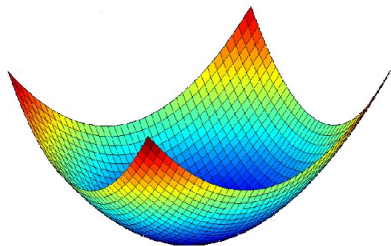
A set $\mathcal{X}$ is convex if for any two points $\mathbf{x}_1, \mathbf{x}_2 \in \mathcal{X}$ and any scalar $\alpha \in [0, 1]$:

$$\alpha \mathbf{x}_2 + (1 - \alpha) \mathbf{x}_1 \in \mathcal{X}$$

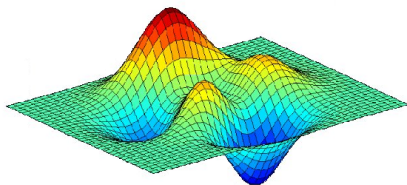
Convex Functions



Convex Functions



convex function



non-convex function

Examples of Convex Functions

Examples of convex functions (when defined over convex sets):

- Constant functions $f(\mathbf{x}) = c$
- Linear functions $f(\mathbf{x}) = \mathbf{x}\mathbf{b} + c$
- Quadratic functions that are positive semi-definite:
 $f(\mathbf{x}) = \mathbf{x}^T \mathbf{A} \mathbf{x} + \mathbf{x}\mathbf{b} + c$
- The absolute value function $f(x) = |x|$
- The ℓ_1 norm $f(\mathbf{x}) = \|\mathbf{x}\|_1$
- The log loss function $f(x) = \log(1 + \exp(-x))$

Algebra of Convex Functions

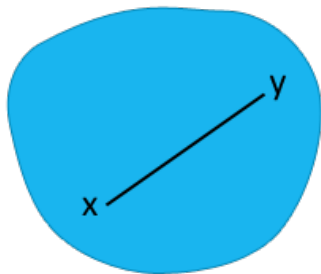
Let $f_1(\mathbf{x}), \dots, f_k(\mathbf{x})$ be convex functions. We have the following composition results:

- A sum $h(\mathbf{x}) = f_1(\mathbf{x}) + \dots + f_k(\mathbf{x})$ of convex functions is a convex function
- A weighted sum $h(\mathbf{x}) = w_1 f_1(\mathbf{x}) + \dots + w_k f_k(\mathbf{x})$ of convex functions with non-negative weights w_i is a convex function.
- The pointwise maximum $h(\mathbf{x}) = \max(f_1(\mathbf{x}), \dots, f_k(\mathbf{x}))$ of a set of convex functions is a convex function
- If $f : \mathbb{R}^M \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ are convex functions and g is non-decreasing, then $h(\mathbf{x}) = g(f(\mathbf{x}))$ is a convex function.

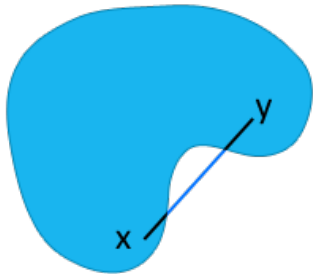
Convex Sets

$$\alpha \mathbf{x}_2 + (1 - \alpha) \mathbf{x}_1 \in \mathcal{X}$$

Convex set



Non - convex set



Convex Sets

Useful examples of convex sets:

- $\mathbb{R}^M$
- $[0, 1]^M$
- The M -dimensional half space $\{\mathbf{x} \mid \mathbf{x}\mathbf{w} + b \leq 0\}$
- The probability simplex of order M : $\{\mathbf{x} \mid \forall i x_i \geq 0, \sum_{i=1}^M x_i = 1\}$
- The ℓ_1 ball $\{\mathbf{x} \mid \mathbf{x} \in \mathbb{R}^M, \|\mathbf{x}\|_1 \leq r\}$
- The ℓ_2 ball $\{\mathbf{x} \mid \mathbf{x} \in \mathbb{R}^M, \|\mathbf{x}\|_2 \leq r\}$
- The set of positive semi-definite matrices
- The intersection of two convex sets

Convex Optimization

A convex optimization problem in standard form consists of three primary components:

$$\min_{\mathbf{x} \in \mathcal{X}} f(\mathbf{x})$$

- Optimization variables: $\mathbf{x} = [x_1, \dots, x_M]$
- Convex objective Function: $f: \mathcal{X} \rightarrow \mathbb{R}$
- Convex feasible set: $\mathcal{X}$

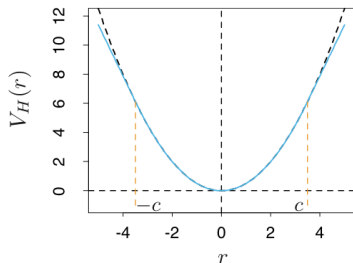
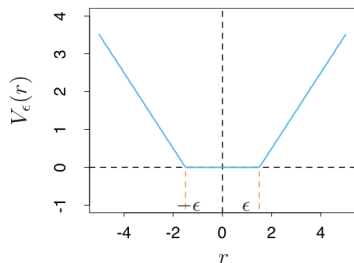
Note: If the function $-f(\mathbf{x})$ is convex, then the function $f(\mathbf{x})$ is called *concave*.

Motivation

- The basic squared loss linear regression model we introduced previously is often referred to as ordinary least squares (OLS) linear regression.
- One of the drawbacks of OLS is that it can be very sensitive to the presence of *outliers*.
- An outlier is a data point with an output value that is much larger or smaller than the output values of other nearby data points.
- Outliers can be caused by errors in the data collection, or by a data generating process that has heavy tails.
- OLS prefers to have many small errors instead of a few large errors, and will thus deflect the regression surface toward outliers in order to minimize the MSE.

Regression with Other Losses

Common alternative convex losses for regression include the epsilon insensitive loss and the Huber loss.



Both of these losses are specifically designed to limit the influence of *outliers* on the model fit.

The specific combination of the epsilon insensitive loss with a squared ℓ_2 norm regularizer is referred to as *support vector regression* (SVR).

Support Vector Regression

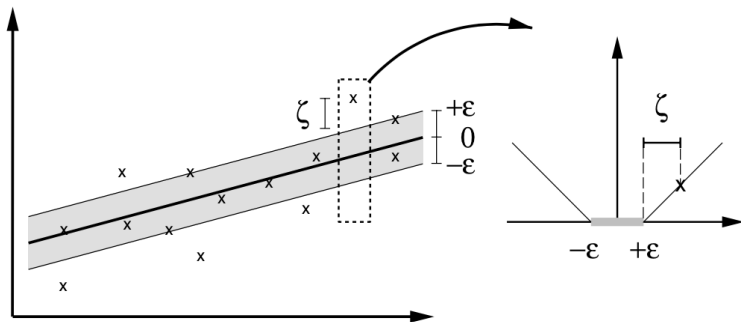
$$f_{\theta}(\mathbf{x}) = \mathbf{x}\theta = \mathbf{x}\mathbf{w} + b$$

$$\hat{\theta} = \arg \min_{\theta} \left(C \sum_{n=1}^N L_{\epsilon}(y_n, f_{\theta}(\mathbf{x}_n)) \right) + \|\mathbf{w}\|_2^2$$

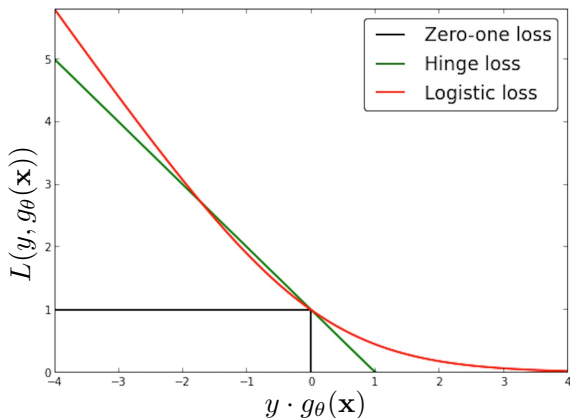
$$L_{\epsilon}(y, y') = \begin{cases} 0 & \dots \text{ if } |y - y'| < \epsilon \\ |y - y'| - \epsilon & \dots \text{ otherwise} \end{cases}$$

Note that for SVR, the objective function is continuous and convex, but not differentiable, thus our optimization toolkit will need updating to fit this model class.

Support Vector Regression Example



Classification Losses



The specific combination of the hinge loss with a squared ℓ_2 norm regularizer is referred to as a *support vector classifier*.

Hinge Loss

- Like the logistic loss, the hinge loss provides an upper bound on the classification error rate. The basic structure of the function is also similar to that of the logistic loss:

$$L_h(y, g_\theta(\mathbf{x})) = \max(0, 1 - yg_\theta(\mathbf{x}))$$

- However, while the hinge loss is continuous and convex, it is not differentiable.

Support Vector Classifier

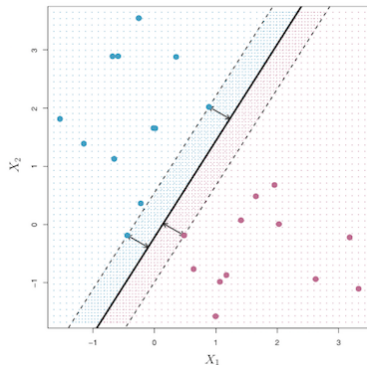
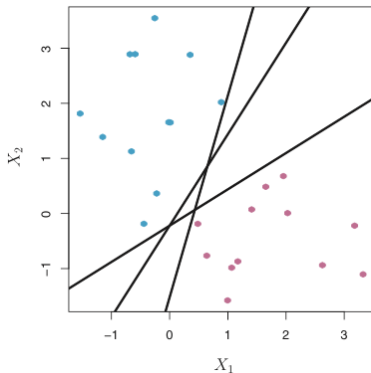
$$f_{\theta}(\mathbf{x}) = \text{sign}(g_{\theta}(\mathbf{x}))$$

$$g_{\theta}(\mathbf{x}) = \mathbf{x}\theta = \mathbf{x}\mathbf{w} + b$$

$$\hat{\theta} = \arg \min_{\theta} C \sum_{n=1}^N \max(0, 1 - y_n g_{\theta}(\mathbf{x}_n)) + \|\mathbf{w}\|_2^2$$

Maximum Margin Property

Part of relevance of SVMs stems from the fact that the hinge loss results in the *maximum margin* decision boundary when the training cases are linearly separable.



Margin

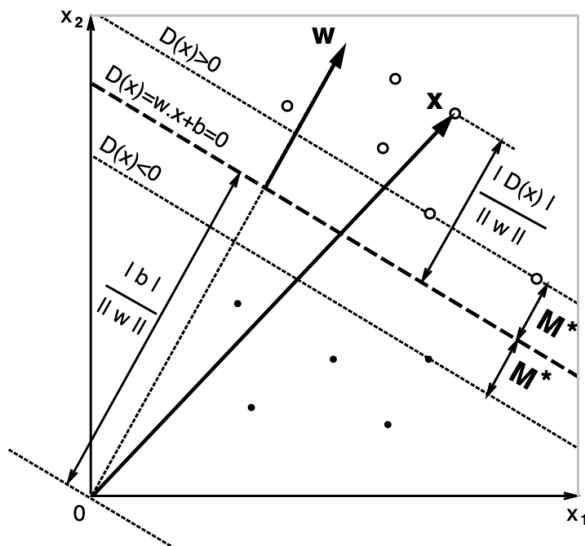
- Suppose we have a data point $\mathbf{x}_n$. The signed distance from the data point to the separating hyperplane $\mathcal{H} = \{\mathbf{x} | g_\theta(\mathbf{x}) = 0\}$ is given by:

$$D_n(\theta) = \frac{y_n g_\theta(\mathbf{x}_n)}{\|\mathbf{w}\|_2}$$

- The value of the margin is then the minimum over the data set of the signed distances of the data points to the separating hyper-plane:

$$M(\theta) = \min_n D_n(\theta)$$

Margin Example



Margin Maximization

The original SVC optimization problem was to maximize the value of the margin as a function of the model parameters. This gives us:

$$\hat{\theta} = \arg \max_{\theta} \min_n D_n(\theta)$$

$$\hat{\theta} = \arg \max_{\theta} \min_n \frac{y_n g_{\theta}(\mathbf{x}_n)}{\|\mathbf{w}\|_2}$$

Quadratic Program Formulation

The direct margin maximization problem is not easy to solve, however, we can re-formulate it into a *quadratic program* for which there are existing solvers.

$$\begin{aligned}\hat{\theta} = \arg \min_{\theta, \epsilon_{1:N}} & \|\mathbf{w}\|_2^2 + C \sum_{n=1}^N \epsilon_n \\ \text{s.t. } & \forall n \ y_n g_{\theta}(\mathbf{x}_n) \geq 1 - \epsilon_n \\ & \forall n \ \epsilon_n \geq 0\end{aligned}$$

The hinge loss formulation of SVC was in fact derived from this formulation of the model.

Support Vector Property

In the linearly separable case, some data points will always fall exactly on the margins. These points are called *support vectors* and they *uniquely determine* the optimal model parameters.

